

MRC Lecture 5 — Derivation Guide 3

Emergent Bilinearity: The Antibody-Antigen Case

Sai T. Reddy — ETH Zurich / BIIE

1. Motivation

WB1 (SantaLucia) and WB2 (PWM) demonstrated **given bilinearity**: the binding energy decomposes as a dot product in raw sequence space, with physical or tabulated parameters as the “query” vector and positional features as the “key” vector. For nucleic acid duplex formation and transcription factor-DNA binding, the physics hands us the decomposition directly.

For protein-protein interfaces — antibody-antigen, T-cell receptor-pMHC, enzyme-substrate — this does not work. The binding energy cannot be written as a positionally independent sum in raw sequence coordinates. Protein binding involves 3D shape complementarity, electrostatic interactions across a geometrically complex interface, hydrophobic packing, and water-mediated contacts. None of these decompose cleanly by sequence position.

This guide derives **why** protein binding resists the raw-sequence decomposition, shows that the bilinear decomposition nevertheless exists in an appropriate learned embedding space (**emergent bilinearity**), and explains why this distinction predicts measurable differences in SFM training behavior — differences the vibe-coding paper observes empirically across six domains.

2. Why Protein Binding Is Different

The antibody-antigen interface. An antibody recognizes an antigen through its **paratope** — a surface formed by six complementarity-determining region (CDR) loops: CDRH1, H2, H3, CDRL1, L2, L3. The antigen’s **epitope** is the corresponding surface on the antigen. The paratope-epitope interface typically involves 15–25 residue-residue contacts with a total buried surface area of ~1,500–2,000 Å².

The contact geometry problem. Consider an antibody with 110 heavy-chain residues plus 107 light-chain residues (217 total), binding to an antigen of 200 residues. The antibody-antigen interface involves ~20 specific contacts. Which 20? This depends on:

- The 3D structure of the antibody (determined by sequence + folding)
- The 3D structure of the antigen (same)
- The relative orientation of the two molecules when bound

A sequence-based feature representation cannot tell you which residues are in contact without first solving the 3D structure. The $L^2 = 43,400$ possible antibody-antigen residue pairs collapse to ~20 actual contacts, but which 20 depends entirely on 3D geometry.

The energy composition problem. Even if we knew the ~20 contacts, the binding energy at each contact is not a simple function of the two residue identities. Contact energies depend on:

- **Residue orientation** (side-chain rotamer, which depends on local structure)
- **Water accessibility** (surface vs. buried contact)
- **Nearby residue identity** (cooperative effects, e.g., a salt bridge’s strength depends on local dielectric)
- **Backbone flexibility** (conformational entropy contributions)

These effects are **non-additive** in raw sequence coordinates. A classic result in antibody engineering: mutations at interface residues show strong epistasis — the effect of mutation $A \rightarrow B$ at residue i depends on the identity of residues j, k, l elsewhere in the interface.

Contrast with DNA duplex formation. DNA duplex formation is approximately positionally independent because adjacent base pairs stack along a fixed geometry (the B-form double helix) with well-characterized nearest-neighbor interactions. The “3D structure” of a DNA duplex is essentially

fixed by sequence — there are no alternative conformations that affect binding. This is why the SantaLucia model works with just 10 parameters.

Protein-protein binding lacks both the fixed geometry (interfaces can adopt many shapes) and the positional independence (distant residues affect local contacts). Hence the need for a different approach.

3. The Bilinearity IS There — Just Not in Sequence Space

Claim. The binding energy for protein-protein interfaces decomposes bilinearly in an appropriate **learned feature space**, even though it does not decompose in raw sequence space.

Formal statement. For an agent protein A and target protein T , there exist encoders Encoder_A and Encoder_T (mapping amino-acid sequences to d -dimensional vectors) such that:

$$\Delta G(A, T) \approx -\frac{1}{\tau} \cdot \text{Encoder}_A(A) \cdot \text{Encoder}_T(T) + C$$

where τ is a learned temperature and C is a constant. **The decomposition exists** — but it lives in the learned embedding space, not in sequence space.

Why this works. The encoder’s job is to project sequence into a representation where the relevant contact physics is linearized. Concretely:

1. The encoder learns to identify which residues are **interface residues** — solvent-exposed, in CDR loops, at the binding site.
2. The encoder weights these residues more heavily in the pooled embedding.
3. The pooled embedding captures the “relevant” features for binding: local chemical environment, charge, hydrophobicity, shape contribution.
4. The dot product between two such embeddings approximates the sum-over-contacts binding energy.

The historical context. This is not a coincidence — it is exactly what protein language models (ESM-2, AntiBERTy, ProtBERT, and others) were trained to do. Their self-supervised training objective (masked language modeling) forces the model to learn representations where structural information is encoded in the embeddings. AlphaFold2 (Jumper et al., 2021) and ESMFold (Lin et al., 2023) demonstrate that these representations contain enough structural information to predict full 3D structures. For binding prediction, we don’t need full structure — we need enough information to compute approximate interface features, which the encoder provides.

4. The Encoder as a Physics Parameterizer

Setup. Consider training an SFM for antibody-antigen binding (the CALM architecture):

$$\begin{aligned} q_i &= \text{Pool}(\text{AntiBERTy}(A_i)) \\ k_j &= \text{Pool}(\text{ESM-2}(T_j)) \\ s_{ij} &= q_i \cdot k_j / \tau \\ P(A_i \text{ binds } T_j) &= \frac{\exp(s_{ij})}{\sum_k \exp(s_{ik})} \end{aligned}$$

Parameters to learn:

- AntiBERTy encoder weights (~600M parameters, but pretrained and often frozen or partially frozen)
- ESM-2 encoder weights (~150M parameters for ESM-2-650M, similarly pretrained)
- Projection heads mapping encoder outputs to shared embedding space (~1M parameters)
- Learned temperature τ (1 scalar)
- Pooling strategy (typically mean or attention-weighted, 0–10k parameters)

The “parameter estimation” framing. Following Reddy (2026b) and the §4.5 analysis in the vibecoding paper: the SFM training is NOT searching a vast function space. The function form is

prescribed by the convergence equation (Lecture 6) — softmax of dot products. Training is estimating the parameters that place each agent and target at the right position in the shared embedding space. This is much more similar to **parameter estimation in a known model** (like fitting Michaelis-Menten parameters to kinetic data) than to **function approximation in an unknown model** (like fitting an arbitrary polynomial to data of unknown form).

The pretrained encoders have already done the hardest work — learning the sequence-to-representation mapping. SFM training then only has to:

1. Align the two encoders' embeddings into a shared space (via projection heads).
2. Tune the temperature to the appropriate scale.
3. Fine-tune encoder weights if needed for the domain specifically.

This is why SFMs work with ~1,000 to ~200,000 training pairs — a fraction of the data needed for scratch-trained models. CALM achieved retrieval success with ~4,000 antibody-antigen pairs; scratch training a comparable model would require orders of magnitude more.

5. Worked Example: Ab-Ag Binding Score

Setup. Take two antibody-antigen pairs: (Ab_1, Ag_1) — a known binder — and (Ab_1, Ag_2) — a known non-binder. Both antibody and antigen sequences are passed through the pretrained encoders:

- AntiBERTy: antibody sequence $\rightarrow$ pooled vector $q_1 \in \mathbb{R}^{768}$
- ESM-2: antigen sequence $\rightarrow$ pooled vector $k_j \in \mathbb{R}^{768}$ for each antigen

Conceptual computation. Suppose (with dummy numerical values):

- $q_1 = (0.3, 0.1, -0.2, \dots)$ (768-dim)
- $k_1 = (0.2, 0.15, -0.1, \dots)$ — true epitope for Ab_1
- $k_2 = (-0.1, 0.05, 0.3, \dots)$ — unrelated antigen

Hypothetical dot products:

$$q_1 \cdot k_1 = 8.5 \text{ (high compatibility)}$$

$$q_1 \cdot k_2 = 1.2 \text{ (low compatibility)}$$

With learned temperature $\tau = 0.5$:

$$s_{11} = 8.5 / 0.5 = 17 \Rightarrow \exp(17) \approx 2.4 \times 10^7$$

$$s_{12} = 1.2 / 0.5 = 2.4 \Rightarrow \exp(2.4) \approx 11$$

For a batch with just these two antigens:

$$P(Ab_1 \rightarrow Ag_1) = \frac{2.4 \times 10^7}{2.4 \times 10^7 + 11} \approx 1.0000$$

$$P(Ab_1 \rightarrow Ag_2) = \frac{11}{2.4 \times 10^7 + 11} \approx 0.00000046$$

Interpretation. The embedding space separates binders from non-binders by a large margin. The dot product operation is the same as in the PWM case — but now the feature vectors are learned 768-dimensional embeddings rather than interpretable 4L-dimensional one-hot encodings. The mathematical operation is identical; the meaning of the features is different.

6. The Parameter Estimation Framing

The claim from Lecture 8 and the vibe-coding paper. SFM training resembles **parameter estimation in a known governing equation**, not **function approximation in an unknown function class**. This framing has empirical predictions:

1. **Fast convergence.** Once the parameters are identified by the data, training peaks early. Function approximation requires extensive search; parameter estimation does not. The vibe-coding paper reports best-epoch fractions of 3–40% of the training budget in four of six domains.

2. **Inverted train-test gap.** At the best checkpoint, validation accuracy exceeds training accuracy in several domains. This is essentially impossible in standard function approximation (where the model is optimized on training data), but is exactly what happens when parameters are pinned by the data: once the parameters are fit, they generalize to held-out data that shares the governing equation.
3. **Out-of-distribution robustness.** In function approximation, OOD performance typically degrades — the learned function is specific to the training distribution. In parameter estimation, the governing equation holds across the distribution, so OOD performance is comparable to ID performance. The vibe-coding paper reports non-degradation or improvement at 40% sequence identity OOD in four of six domains.

The Michaelis-Menten analogy. An enzymologist fitting Michaelis-Menten parameters to kinetic data works with ~ 10 data points and obtains V_{max} and K_m that generalize to all substrate concentrations. The parameters are pinned by the data; the function is known. Showing 10,000 data points does not improve the fit beyond tighter confidence intervals. An engineer fitting an arbitrary polynomial to the same data would need thousands of points and would find the fit does not extrapolate.

SFM training with emergent bilinearity is closer to the Michaelis-Menten case — the governing equation (convergence equation) is known, and the parameters are the embedding coordinates. This is why SFMs work with 4,000 pairs where CLIP needed 400M image-text pairs.

7. The Level 3 Claim

Level 1 alignment: A Level 1 system implements the governing equation of a physical process as a **computational shortcut**. Example: McCulloch-Pitts neurons implement integrate-and-fire as a step function approximation.

Level 2 alignment: A Level 2 system's architecture is **inspired by** the physical process but does not implement the governing equation. Example: convolutional neural networks use local connectivity inspired by visual cortex receptive fields but do not implement the actual neuroscience.

Level 3 alignment: A Level 3 system's architecture **IS** the governing equation of the physical process, with the parameters of the architecture identified with the parameters of the equation.

SFMs are Level 3 for molecular recognition.

- The convergence equation says: $P(\text{bind}) = \text{softmax}(q \cdot k / \tau)$
- The SFM computes: $P(\text{retrieve}) = \text{softmax}(q \cdot k / \tau)$
- Same equation. Parameters are learnable; the equation is not.

This is what makes emergent bilinearity compelling even though the decomposition is not in raw sequence space: the decomposition **IS** the governing equation, just in a different basis. The encoder changes basis; the equation is invariant.

8. Predictions for Training Dynamics

Based on the emergent vs. given distinction, we predict:

Given bilinearity (miR-SFM, tSFM, crisprSFM):

- **Smaller effective parameter count** — the form is prescribed in raw coordinates, so fewer parameters need to be fit.
- **Faster convergence** — less search space to traverse.
- **Tighter calibration** — embedding distances correspond to physical energies with minimal unit conversion.
- **Smaller D_c** — the scaling constant governing how much data is needed.

Emergent bilinearity (CALM, TCR-SFM, eSFM, dtSFM, mhcSFM):

- **Larger effective parameter count** — encoder must discover the right representation.
- **Slower convergence** — more parameters to fit (though pretraining compresses this).

- **Looser calibration** — additional unit conversion from embedding space to energy.
- **Larger D_c** — more data needed.
- **But still Level 3** — the governing equation is still prescribed; only the basis changes.

Empirical confirmation from the vibe-coding paper:

Domain	Bilinearity	Best epoch	R@1 (ID)
miR-SFM	Given	8–18%	98.4%
mhcSFM	Emergent	12–18%	65–93%
eSFM	Emergent	15–40%	62–86%
dtSFM v2	Emergent	3–8%	28–61%
crisprSF	Given*	52–99%	95%
M			
tSFM	Given*	42–98%	15–37%

*Given with competing trivial baseline

The caveat for crisprSFM and tSFM. Both are given-bilinearity systems but show slow in-distribution convergence. The vibe-coding paper’s explanation: a competing trivial baseline exists in their loss landscape (Hamming distance for CRISPR; sequence homology for TF) that already solves the in-distribution problem to first order. The SFM must spend many epochs learning features that exceed this baseline. Consistent with this explanation, their OOD convergence IS fast (1–38 epochs at OOD-80% splits) — once the trivial baseline no longer applies, the model behaves as predicted for given bilinearity.

9. Why This Matters for Your SFM Project

For students choosing a domain (Lectures 9–11):

If you pick a given-bilinearity domain (tSFM, miR-SFM, crisprSFM):

- Expect fast in-distribution convergence (best epoch 10–30% of budget typical)
- Achievable R@1 > 90% with modest data (~20,000–50,000 pairs) for miRNA-like cases
- Watch for competing trivial baselines (Hamming, seed matching) that may slow ID convergence
- Validation against biophysical baselines (TargetScan, Cas-OFFinder) is a strong evaluation

If you pick an emergent-bilinearity domain (CALM-style Ab-Ag, TCR-pMHC, pMHC):

- Expect slower convergence (best epoch 10–40% of budget)
- Need more data (~50,000–200,000 pairs) for strong results
- Pretrained encoders (ESM-2, AntiBERTy) do the heavy lifting — fine-tune carefully
- Validation against experimental data from your own lab is feasible (your own antibody libraries, TCR repertoire, patient HLA haplotypes)
- OOD evaluation via MMseqs2 sequence-identity splits at 80%/60%/40% is informative

The architecture is the same in both cases — the only changes are encoder selection, data source, and hard negative strategy. The core training loop, loss function, and softmax operation are identical. If you’re not modifying the core — you’re doing it right.

10. Second-Order Corrections to Emergent Bilinearity

The dot-product approximation in embedding space is approximate. Known deviations:

Allosteric effects. Binding at one site can affect binding at another through conformational changes. The pure dot product assumes independent contributions; allostery violates this. Architectural fix: cross-attention layers that allow the agent and target representations to interact before the final dot product.

Concerted conformational changes. Some antibodies undergo significant CDR reorientation upon antigen binding. A fixed embedding cannot capture “what the antibody looks like when bound.” Architectural fix: equivariant encoders that can be conditioned on the partner.

Coupled water networks. Interfaces often contain structured water molecules that mediate binding and are difficult to capture from sequence alone. Architectural fix: explicit representation of interfacial solvation (out of scope for current SFMs).

Kinetic trapping. Some binding events are not in thermodynamic equilibrium (irreversible covalent modification, conformational trapping). The Boltzmann framework assumes equilibrium; non-equilibrium systems require explicit kinetic modeling. Architectural fix: not possible with the base SFM — requires additional kinetic inputs.

In practice: for retrieval tasks (which target does this agent bind?), the dot-product approximation is almost always sufficient — the dominant binding terms suffice to rank candidates correctly. For absolute affinity prediction at the tails (pM tight binders, mM weak binders), deviations can matter. The affinity calibration framework (future work, Reddy 2026b §S4) tests for these deviations via residual analysis.

11. Summary

Concept	Result	Significance
The sequence-space problem	Protein contacts depend on 3D structure	No naive dot-product in raw sequence
The embedding-space solution	$\Delta G \approx -q \cdot k / \tau + C$	Learned encoder provides the decomposition
Encoder role	Parameterize the physics from data	ESM-2, AntiBERTy do the heavy lifting
Convergence equation	Still applies universally	Same softmax, same architecture
Level 3 alignment	Architecture IS the governing equation	Parameter estimation, not function approximation
Training dynamics	Slower than given, but still Level 3	Fast convergence, inverted gap, OOD robustness
Your project	Same architecture, different encoders	Code is the same; domain determines dynamics

12. Checkpoint Questions

Q1. A student argues: “If the decomposition is emergent — only in embedding space — then SFMs aren’t really Level 3 aligned. They’re Level 2 at best. Level 3 should require the equation to hold in raw coordinates.”

Answer: The student confuses “decomposition in a specific basis” with “alignment with the governing equation.” The convergence equation says: $P(\text{bind}) = \text{softmax}(q \cdot k / \tau)$. It does NOT say the dot product must be in sequence coordinates. The equation requires bilinearity, but bilinearity is a basis-independent property — it holds in one basis iff it holds in every basis related by an invertible linear transformation. For given-bilinearity systems (DNA), the raw sequence basis is already a convenient bilinearity basis. For emergent systems (protein), a learned basis is required. Both implement the same equation; both are Level 3. The encoder is a change of basis, not a change of equation. (This is analogous to: rotating to a different coordinate system in classical mechanics does not change Newton’s laws; it just changes their coordinate expression.)

Q2. Why don’t we see emergent bilinearity for DNA-DNA duplex formation? Shouldn’t a neural network on DNA sequences also work?

Answer: It would work — but it would be overcomplicated. For a system where the raw-sequence dot product already captures the physics to within 0.5 kcal/mol accuracy (SantaLucia), training a neural network adds parameters and data requirements for no accuracy gain. The given-bilinearity systems have a principled reason to use simpler models: the physics already gives you what you need. The emergent-bilinearity systems need neural networks because raw-sequence features do not capture the physics. A well-designed SFM for miRNA-mRNA (miR-SFM) exploits the given bilinearity in its

architecture choices (e.g., using RNA-FM or DNABERT-2 with appropriate pooling) rather than fighting it.

Q3. The vibe-coding paper reports that dtSFM v2 has the fastest convergence (3–8% of budget) among the six SFMs despite being emergent bilinearity. How is this consistent with the given/emergent predictions?

Answer: Several factors: 1. The pretrained encoders (MoLFormer-XL for drugs, ESM-2 for proteins) are very mature and extensively trained on large domains. They provide high-quality representations that require minimal fine-tuning. 2. Drug-target binding has well-defined binding pockets that strongly constrain which residues matter. The encoder doesn't have to learn about the full protein — just the pocket region, which is relatively conserved structurally. 3. v2 specifically used a much more diverse dataset than v1 (3.2× proteins, 5.1× drugs), improving parameter identifiability. The prediction is that emergent bilinearity has LARGER D_c and MORE parameters to fit than given bilinearity, requiring relatively more data. But given high-quality pretrained encoders and identifiable parameters, the convergence CAN be fast. The physics signature (fast convergence, inverted gap, OOD robustness) appears in all four fast-converging domains (miR-SFM, mhcSFM, eSFM, dtSFM v2), consistent with all of them being Level 3.

Q4. For the students' SFM projects, how should the “competing trivial baseline” caveat affect domain choice?

Answer: If you're picking a domain for a clean demonstration of the framework, avoid domains where a simple sequence-similarity baseline already solves the in-distribution problem to first order. These domains (CRISPR off-targets via Hamming; TF binding via sequence homology) will show slow in-distribution convergence that may look discouraging in a short-timeline project. Instead, prefer domains with: - Clear biological signal (binding is well-defined by physics) - Limited trivial baselines (raw sequence similarity doesn't work) - Good pretrained encoders (ESM-2, DNABERT-2, MoLFormer, RNA-FM)

Good choices: miR-SFM, CALM-style Ab-Ag, TCR-pMHC, pMHC (mhcSFM). Each has these properties. If you do pick a trivial-baseline-dominated domain, design the evaluation to include OOD splits where the trivial baseline fails — that's where the SFM will shine.

13. Connection to Future Lectures

- **Lecture 6:** Luce's uniqueness theorem proves that the softmax form applied to these embedding-space dot products is the unique function satisfying five axioms of competitive selection. The emergent-vs-given distinction is orthogonal to uniqueness — both yield the same functional form, and both inherit the uniqueness.
- **Lecture 7:** The full SFM architecture is derived. For emergent-bilinearity domains, the derivation explicitly includes the learned encoders (ESM-2, AntiBERTy) and projection heads as the basis-change operation.
- **Lecture 8:** The Level 1/2/3 classification is developed formally. SFMs are Level 3 across both given and emergent bilinearity. The six prototyped domains span both cases with a single architecture — the strongest empirical evidence for Level 3 alignment.
- **Lectures 9–11:** Students building SFMs will experience both regimes firsthand. Domain choice determines training dynamics (as described here), but the code is the same. The results workshop (Lecture 11) will reveal the empirical signatures — fast convergence, inverted gaps, OOD behavior — in the students' own training logs.
- **Lecture 12:** The failure conditions include “emergent bilinearity approximation breaks down at specific affinity extremes.” This is a testable prediction, not a framework-breaking failure — second-order corrections exist and map to known architectural variants.